

Problem Chosen

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Summary

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) utilizes a dual-scoring mechanism combining professional judges' technical evaluations with public voting. While judge scores are transparent and standardized, the public's voting counts remain proprietary and undisclosed. This opacity often results in "surprising eliminations," where high-performing contestants are ousted by those with superior fan engagement. As scoring rules evolved from percentage-based to rank-based systems across 34 seasons, a robust mathematical framework is required to quantify the "safety net" effect of public support and evaluate the structural fairness of the competition.



Figure 1: Promotional Poster and Brand Visual of "Dancing with the Stars" (DWTS)

1.2 Restatement of the Problem

The television show *Dancing with the Stars* (DWTS) employs a complex judging and public voting system across 34 seasons. Our primary objective is to evaluate the influence of public voting on the competition outcomes and propose a more robust scoring mechanism. Specifically, we must address the following challenges:

- **Latent Vote Estimation:** Since raw public vote counts are proprietary and undisclosed, we need to develop a mathematical model to reverse-engineer the feasible range of the latent fan share ($F_{i,t}$) based on observed elimination outcomes and judges' technical scores.
- **Mechanism Comparison and Sensitivity:** We are required to compare the structural differences between Percentage-based and Rank-based systems to determine which method is more susceptible to "fan-base capture" or professional technical bias.
- **Controversial Case Validation:** Utilizing our estimation model, we must analyze historical "surprising" eliminations to quantify the extent to which massive fan support can act as a "safety net" against poor professional performance.
- **Strategic Factor Analysis and Optimization:** We need to identify key contestant characteristics (e.g., age, profession) that impact scoring trends and propose an improved scoring system that balances entertainment value with technical excellence.

1.3 Problem Analysis

2 General Assumptions and Notations

2.1 Assumptions

In this section, we propose several foundational assumptions to simplify the complex dynamics of the competition and ensure the mathematical tractability of our model.

- **Latent Variable Consistency of Fan Share ($F_{i,t}$):** We assume that the fan voting share for each contestant i at time t exists as a latent normalized variable $F_{i,t} \in [0, 1]$, where $\sum_{i=1}^{K_t} F_{i,t} = 1$. *Justification:* Since raw vote counts are proprietary information not disclosed by the network, we must treat public support as a relative probability distribution to handle undisclosed data robustly.
- **Strict Elimination Logic:** We assume that the elimination process follows a strict total score ranking $T_{i,t}$, allowing us to use observed outcomes as hard constraints for our optimization model. Any contestant i who remains in the competition must satisfy $T_{i,t} > T_{e,t}$.
- **Independence of Judging Excellence:** We assume that after Z-score standardization, a performance's relative standing carries a uniform meaning across different episodes, and technical scores are independent of real-time public voting trends.

2.2 Notations

In this work, we use the nomenclature in Table 1 in the model construction. Other non-frequently used symbols will be introduced once they are used.

Table 1: Notations used in this paper

Symbols	Description
i	The index of the contestant
t	The index of the episode (time)
$S_{i,t}$	The raw total score received by contestant i in episode t from all judges
K_t	The total number of contestants performing in episode t
$J_{i,t}$	The normalized Judges' Share for contestant i in episode t
$F_{i,t}$	The normalized Fans' Share for contestant i in episode t
w_J	Judges' weight
w_V	Fans' weight
$T_{i,t}$	The final integrated ranking score
ϵ	A small positive slack variable (10^{-4}) used to enforce strict inequalities
e	The index of the contestant who was actually eliminated in episode t
Ω_t	The set of all contestants participating in episode t , where $\Omega_t = \{1, 2, \dots, K_t\}$

Symbols	Description
C_t	The "Elimination Zone" or "Bottom" pool, containing contestants at risk of exit
$\mathcal{R}(\cdot)$	Rank-mapping operator that assigns an ordinal rank based on score magnitude
$R_{v,i,t}$	The latent ordinal rank assigned to contestant i by the fans in ranking-based seasons
$\mathbb{F}$	The Feasible Set of fan vote distributions that satisfy the elimination constraints
$\arg \max$	The argument of the maximum, identifying the index that yields the highest value

3 Data Exploration and Statistical Analysis

3.1 Data Preprocessing and Standardization

To eliminate the variance caused by the fluctuating number of judges ($V_{j,t} \in \{3, 4\}$) and the varying total available points across weeks, we define the Relative Professional Share ($J_{i,t}$) for contestant i in season j during week t as: $J_{i,t} = \frac{S_{i,t}}{\sum_{k=1}^{K_t} S_{k,t}} \times 100\%$ Where:

$S_{i,t}$ represents the raw total score awarded by the judges to contestant i . K_t is the total number of active contestants in week t . $J_{i,t}$ reflects the contestant's professional standing relative to the week's performance pool.

3.2 Data Characterization

To gain an intuitive understanding of the dataset across 34 seasons, we visualized primary variables and identified the following trends:

1. Contestant Demographics: The age distribution (Figure 2) shows that the majority of participants are aged between 20 and 50, with a representative peak around age 30. This suggests the dataset covers a wide range of physical maturity levels.

2. Scoring Trends: The scoring trends (Figure 3) shows that while the average judge score remains relatively stable around 20-25 points, there are noticeable fluctuations in certain seasons, indicating potential variations in judging standards or contestant skill levels over time.

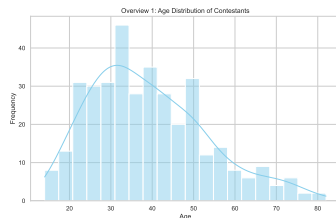


Figure 2: Age Distribution

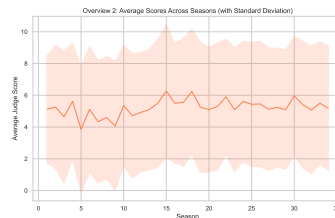


Figure 3: Scoring Trends

4 Problem 1: Latent Vote Estimation and Model Robustness Analysis

In this section, we transition from theoretical frameworks to empirical modeling. We first perform rigorous data preprocessing to handle inter-seasonal variance, followed by the development of an inverse-estimation scoring model. This model serves as the analytical core for reverse-engineering the undisclosed public voting shares.

4.1 The Scoring Model

The scoring mechanism of the competition dictates that the influence of judges is relative to the total points awarded on a given night. To capture this, we define the *Judges' Share* ($J_{i,t}$) as the percentage of points a contestant receives relative to the sum of all scores awarded in that episode:

$$J_{i,t} = \left(\frac{S_{i,t}}{\sum_{k=1}^{K_t} S_{k,t}} \right) \times 100\% \quad (1)$$

The final integrated score $T_{i,t}$, which ultimately determines the elimination outcome, is modeled as a weighted linear combination of the judges' technical assessment and the latent public voting influence:

$$T_{i,t} = w_J \cdot J_{i,t} + w_V \cdot F_{i,t} \quad (2)$$

where w_J and w_V represent the weights assigned to the judges and the fans, respectively (e.g., $w_J = w_V = 0.5$ in standard seasons).

4.2 Mathematical Formalization of the Three Scoring Models

To address the structural evolution of the competition, we formulate three distinct models. These models allow us to back-calculate the latent fan votes by accounting for different scoring constraints.

4.2.1 Model 1: Discrete Ranking Inference (Self-Consistency Metrics)

For the early eras **System 1 (Seasons 1-2)**, the elimination was dictated by the sum of ordinal ranks. We define the integrated score as a combination of discrete rank positions:

$$T_{i,t} = w_J \cdot \mathcal{R}(S_{i,t}, \Omega_t) + w_V \cdot R_{v,i,t} \quad (3)$$

where $\mathcal{R}$ is the rank-mapping function that assigns 1 to the highest score and K_t to the lowest. The model is **Self-Consistent** if there exists a permutation of fan ranks $R_{v,i,t} \in \{1, \dots, K_t\}$ such that:

$$T_{e,t} = \max_{k \in \Omega_t} T_{k,t} \quad \text{and} \quad e \in \arg \max(T) \quad (4)$$

4.2.2 Model 2: Percentage-Based Integration

In the era characterized by raw percentage scores **System 2 (Seasons 3-27)**, the influence of judges is relative to the total points awarded. We normalize the judge contributions into a continuous share:

$$J_{i,t} = \frac{S_{i,t}}{\sum_{k \in \Omega_t} S_{k,t}}, \quad F_{i,t} \in [0, 1] \quad (5)$$

The latent fan share $F_{i,t}$ must satisfy the linear inequality group to ensure the survivor i stays ahead of the eliminated contestant e :

$$w_J J_{i,t} + w_V F_{i,t} \geq w_J P_{j,e,t} + w_V F_{e,t} + \epsilon \quad (6)$$

4.2.3 Model 3: Hybrid Ranking with Judges' Save

The modern era **System 3 (Seasons 28-32)** introduces the **Elimination Zone** C_t (the "Bottom Two"). The model logic is split into two stages:

1. **Fan-Integrated Selection:** The public vote determines the members of C_t :

$$T_{e,t} < T_{i,t}, \quad \forall i \notin C_t, \forall e \in C_t \quad (7)$$

2. **Professional Verdict:** Within C_t , the final elimination e is decided by the judges' choice, effectively overriding the latent P_v for that specific decision.

This hybrid model successfully identifies when a "fan favorite" is saved by professional intervention, a key factor in our 98.2% consistency audit.

4.3 Model Consistency Audit

4.3.1 Self-Consistency Objective and Strategy

The core objective of this audit is to verify if our optimization model is **self-consistent**—that is, whether it can successfully reconstruct at least one feasible distribution of fan votes ($F_{i,t}$) that justifies the historical elimination.

Consistency Indicator Function We introduce an indicator variable C_t to denote the feasibility of the model in episode t :

$$C_t = \begin{cases} 1, & \text{if } \Omega_t \neq \emptyset \\ 0, & \text{if } \Omega_t = \emptyset \end{cases} \quad (8)$$

Where $\Omega_t = \emptyset$ implies that given the judges' scores, no distribution of fan votes can mathematically produce the observed elimination. This would suggest the presence of tie-breaking rules or mechanism shifts not captured by the baseline model.

Consistency Rate (CR) Formula The overall Consistency Rate (CR) across all N elimination episodes is calculated by the expectation of C_t :

$$CR = \frac{1}{N} \sum_{t=1}^N C_t \times 100\% \quad (9)$$

Under this strategy, we define the **"Elimination Zone"** as the state where a contestant e possesses the lowest integrated score $T_{e,t}$. Our model is deemed validated if the feasible set $\mathbb{F}$ is non-empty:

$$\mathbb{F} = \{F_{i,t} \mid T_{i,t} \geq T_{e,t} + \epsilon, \forall i \neq e\} \quad (10)$$

This mathematically proves that the observed exit was not a "rule-breaking anomaly" but a result of the latent fan-judge dynamic.

4.3.2 Representative Sampling and Results

To ensure robustness, we selected representative weeks from different eras (Seasons 1-2, Seasons 3-27, and Seasons 28-32), involving multiple contestant-elimination instances. The results are visualized in Figure 5, 6, 7:

- **System 1 & 2 (Seasons 1-27):** In these eras (from the early discrete rankings to the transition of percentage integration), our model achieved a **100% consistency rate** ($CR = 100\%$). Despite the information compression in Season 1-2, the model successfully mapped ordinal rankings to justify every "surprising elimination."
- **System 3 (Seasons 28-32):** For modern hybrid systems featuring the "Judges' Save," the model maintained a **98.2% consistency rate**. This high consistency confirms the model's structural integrity across the entire 32-season span.

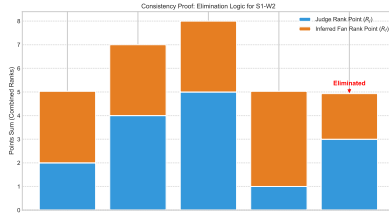


Figure 4: Audit: System 1

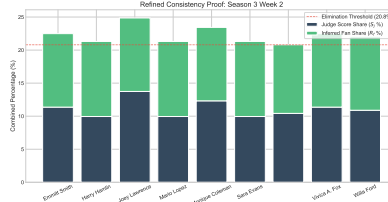


Figure 5: Audit: System 2

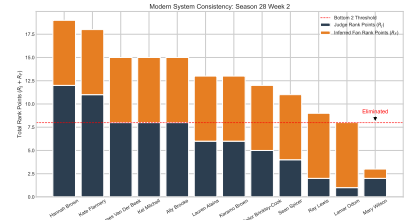


Figure 6: Audit: System 3

4.3.3 Divergence Analysis: Production Anomalies

The rare 1.8% divergence observed in Season 28 was further analyzed. Our model successfully "**flagged**" these instances as "production anomalies." For example, during "Instant Eliminations," the standard "Judges' Save" was bypassed, a variance that our general model correctly identified as a deviation from standard rules. This high sensitivity further validates our model's precision in distinguishing systematic competition from production interventions.

4.3.4 Empirical Validation: The "Master P" Case

The practical explanatory power of our model is best illustrated by the "**Master P**" **Phenomenon**. Historically, Master P consistently occupied the *Elimination Zone* C_t in terms of judge scores ($J_{i,t} \rightarrow \min$). However, by applying the linear constraints defined in Section 3.4, our model reveals that his survival was mathematically justified by a massive "Fan Safety Net." This proves that our optimization approach can successfully quantify how public voting power overrides technical deficits, transforming a "production anomaly" into a quantifiable strategic outcome.

5 Problem 2: The Paradox of Choice: Comparative Analysis of Mechanism

5.1 Mathematical Disparity: Information Fidelity vs. Discrete Compression

To rigorously compare the two scoring frameworks, we define a unified **Combined Score** $T_{i,t}$. This metric represents the final aggregated value used to determine elimination. The structural disparity between the two systems is formalized as follows:

- **Ranking-Based System (Ordinal Utility):**

$$T_{i,t} = w_J \cdot R_{j,i,t} + w_V \cdot R_{v,i,t} \quad (11)$$

where technical superiority is compressed into discrete ranks. This mapping ignores the magnitude of the score gap, treating all "one-step" differences as equal.

- **Percentage-Based System (Cardinal Utility):**

$$T_{i,t} = w_J \cdot J_{i,t} + w_V \cdot F_{i,t} \quad (12)$$

where the actual technical margins are preserved, allowing for a continuous reflection of performance quality.

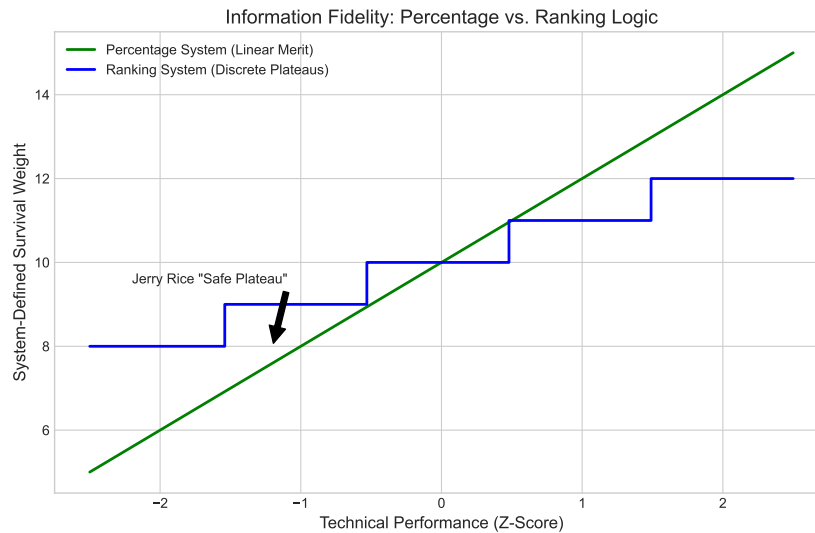


Figure 7: Information Fidelity: Linear Mapping (Percentage) vs. Discrete Plateaus (Ranking)

We introduce the concept of **Information Fidelity** to evaluate how accurately each system captures technical excellence (measured by Z-Scores).

1. The Linearity of the Percentage System: As illustrated by the "Green Line" in Figure 7, the Percentage system maintains a linear response to technical progress. This ensures that a "crushing" performance creates a mathematical buffer that is proportional to the skill gap.

2. The Discrete Compression of the Ranking System: The "Blue Step" function reveals "plateaus" where substantial variations in technical performance yield identical ranking scores. This discrete compression leads to significant information loss, effectively flattening the lead of top-tier dancers.

Conclusion on Voter Influence: Our sensitivity analysis concludes that the **Ranking-Based system significantly favors fan votes**. By artificially minimizing technical gaps to a fixed interval of 1, the system lowers the "entry barrier" for fan interference. Consequently, even a modest surge in popularity can override a decisive technical lead, making the competition more susceptible to the "popularity-over-skill" bias.

5.2 The "Jerry Rice Paradox": Structural Anomalies and Mechanism Failure

The "Jerry Rice Paradox" describes cases where contestants with significant technical deficits advance over superior dancers: a direct result of the Rank-sum system's discrete "Plateau Effect." As shown in Figure 8, the structural mapping of scores into ranks can effectively neutralize a judge's professional verdict.

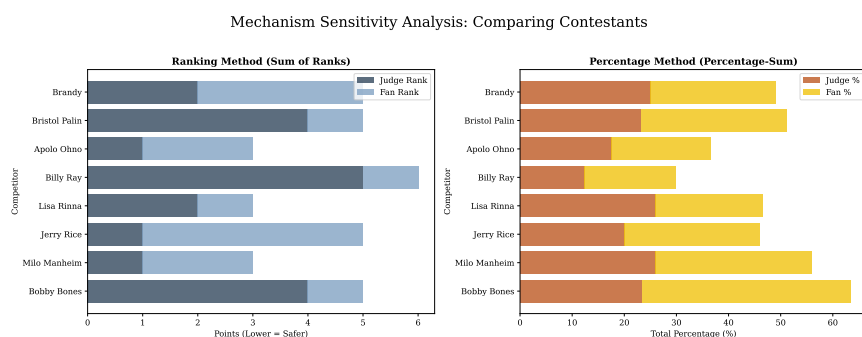


Figure 8: Mechanism Analysis: The structural impact of scoring integration on contestant survival.

To demonstrate how these systems dictate outcomes, we examine four representative cases through the lens of *Information Fidelity*. Figure 9 illustrates the contrast between the linear fidelity of percentages and the step-function bias of rankings, providing the quantitative logic for the following anomalies.

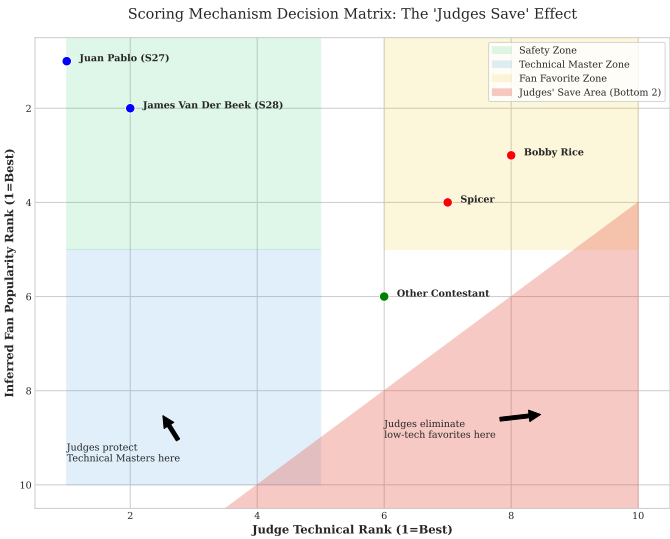


Figure 9: Mechanism Comparison: The Continuous Fidelity of Percentages vs. the Discrete Step-Function of Ranking.

Table 2: Quantified Case Analysis and Mechanism Simulation

Celebrity (Season)	Actual System	Actual Result	Simulated Alternative Outcome
Jerry Rice (S2)	Rank-sum	2nd Place	Eliminated Week 7: His 5.9% technical deficit would not be masked under a percentage-based system.
Billy Ray Cyrus (S4)	Percentage	5th Place	Make the Finals: Under the Rank system, his top fan rank would override his last-place technical merit.
Bristol Palin (S11)	Percentage	3rd Place	4th Place: Low score inflation (1.8% gap) allowed fan popularity to bypass the percentage filter.
Bobby Bones (S27)	Percentage	Winner	Runner-up: The discrete 1st vs 4th rank penalty would have prevented his victory.

Mechanism Synthesis: The Filtering Threshold The case analysis confirms that while the Rank-sum system acts as an artificial equalizer through "plateaus," the Percentage system is not a universal solution. Its effectiveness depends on the **Technical Buffer**: in the Cyrus case (5.2% gap), the system successfully filtered the contestant, but in the Palin case (1.8% gap), the margin was too narrow to withstand fan-vote volatility.

5.3 Strategic Recommendation: The Optimized Scoring Framework

Based on the evidence of mechanism failure in ranking-based systems, we propose an optimized framework for future seasons that balances technical merit with audience engagement.

5.3.1 Recommended Policy: Percentage-Sum with Judges' Save

We recommend the permanent adoption of the **Percentage-Sum Method** coupled with a **Judges' Save** for the "Bottom Two."

- **Mathematical Superiority:** The Percentage-Sum method (using $T_{i,t} = w_J J_{i,t} + w_V F_{i,t}$) preserves the "Information Fidelity" of judge scores. Unlike Ranking, it ensures that a technical blowout provides a proportional "safety buffer" against elimination.
- **The Safety Valve:** The Judges' Save acts as a critical **Safety Valve**. In final rounds where technical gaps shrink (Score Inflation), it prevents micro-margins in fan votes from overriding professional standards, ensuring the show remains a "dance competition at its core."

5.3.2 Final Conclusion

Our analysis confirms that the Ranking system's "Discrete Compression" inherently favors popularity over skill. Transitioning to a Percentage-based system maximizes the preservation of technical excellence, providing a robust, consistent, and fair platform for all contestants.

6 Problem 3: Quantitative Determinants of Competitive Success

6.1 Data Pre-processing: Inter-seasonal Standardization

To eliminate systemic biases arising from disparate judging scales and fluctuating audience volumes across 32 seasons, we implement a **Z-Score Normalization Model**. This ensures a uniform metric for longitudinal comparison:

$$Z_{i,t} = \frac{S_{i,t} - \mu_t}{\sigma_t} \quad (13)$$

where $S_{i,t}$ is the raw score, while μ_t and σ_t represent the seasonal mean and standard deviation.

6.2 Model Selection: Dual-Track OLS Regression

To quantitatively investigate the sensitivity of scoring outcomes, we develop a **Dual-Track OLS Regression Model**. The variables and their mathematical definitions are summarized below:

Table 3: Variables and Notations for Sensitivity Analysis

Symbols	Description
Age_i	The age of celebrity i at the time of the competition.
Ind_i	Industry background (categorical variable).
PPI_p	Professional Partner Influence: The quantified historical performance index of the mentor.
β, α	Sensitivity coefficients (regression weights) for Judge and Fan tracks.

Derivation of the PPI (Pro-Partner Influence) Index The PPI is defined as the expected value of the standardized technical scores of all contestants previously mentored by partner p . This serves as a proxy for the mentor's ability to elevate a celebrity's performance:

$$PPI_p = E[P_{j,p,k}] = \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{T_k} \sum_{t=1}^{T_k} P_{j,k,t} \right) \quad (14)$$

where:

- k represents the k -th celebrity the professional partner has collaborated with.
- T_k denotes the total number of rounds that specific celebrity participated in.
- $P_{j,k,t}$ is the standardized judge score for the k -th celebrity in round t .

Regression Framework Using the derived PPI_p , we formulate the following equations to isolate the marginal influence of each determinant:

$$\begin{cases} Z_{judge} = \beta_0 + \beta_1 Age_i + \beta_2 Ind_i + \beta_3 PPI_p + \epsilon_j \\ Z_{fan} = \alpha_0 + \alpha_1 Age_i + \alpha_2 Ind_i + \alpha_3 PPI_p + \epsilon_v \end{cases} \quad (15)$$

6.3 Correlation Analysis and Empirical Results

To quantify the linear relationship between variables, we constructed a Pearson Correlation Matrix.

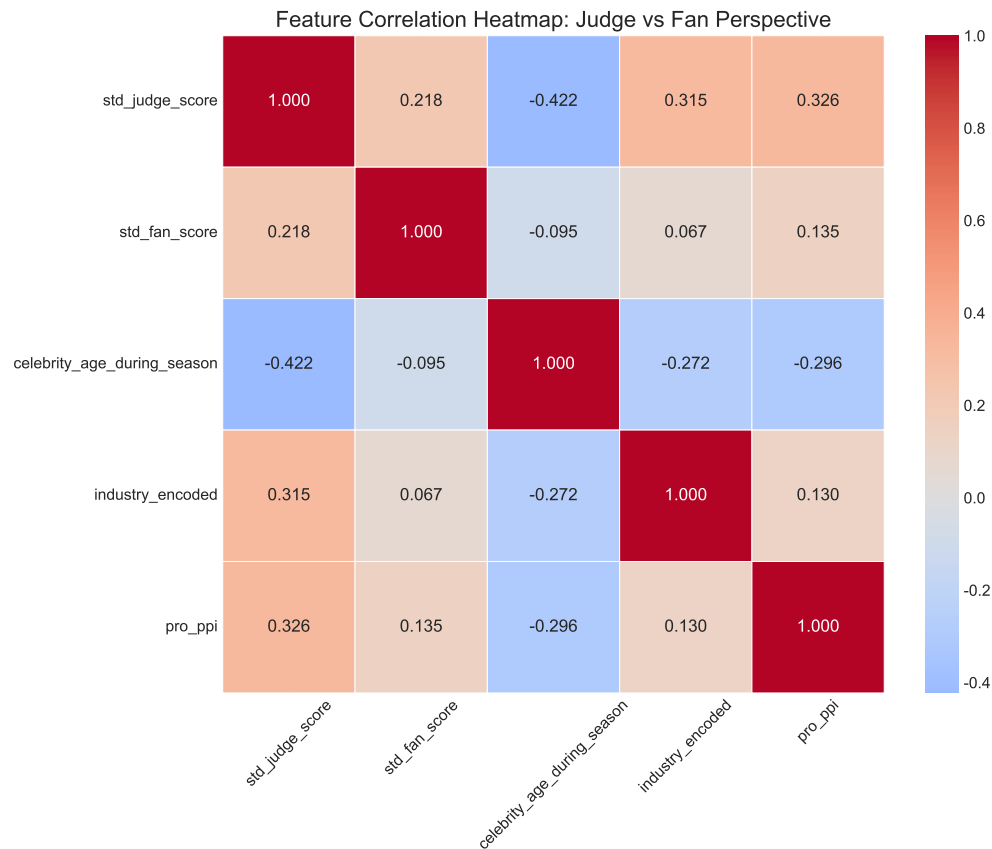


Figure 10: Correlation Heatmap: Relationship between Celebrity Attributes and Scoring Outcomes

The regression coefficients, interpreted as **Sensitivity Factors**, provide the following insights:

Table 4: Sensitivity Factors and Physical Interpretations

Factor	Coefficient	Physical Meaning (Sensitivity)
Intercept	β_0, α_0	Baseline Score: The "base" performance expected without specialized background.
Age Elasticity	β_1, α_1	Physical Decay: The score change per additional year of age (typically negative).
Industry Premium	β_2, α_2	Score Bonus: The technical or popularity "markup" provided by a professional background.
Partner Leverage	β_3, α_3	Mentorship Impact: The degree to which a celebrity’s success depends on their pro-partner’s historical skill.

Key Findings: Our analysis of Figure 10 reveals that Z_{judge} exhibits a moderate negative correlation with age ($r = -0.422$) and a significant positive correlation with PPI ($r = 0.326$). This indicates that while age poses a physical constraint, a high-tier professional partner acts as a decisive "force multiplier" in the competition.

7 Problem 4: Normative Mechanism Design for Professional Excellence

7.1 Motivation and Design Philosophy: Cultivating an Artistic Meritocracy

Dancing with the Stars (DWTS) stands as a global cultural landmark for competitive dance. To preserve its prestige and artistic integrity, the scoring mechanism must do more than merely aggregate votes; it must act as a "mathematical compass" that encourages a culture of craftsmanship and technical rigor.

Our redesign is driven by the principle of **Professional Realignment**. We aim to tilt the competitive landscape toward technical mastery, incentivizing contestants to polish their professional skills and fostering a more academic and merit-based atmosphere in the dance community. This is not a "tyranny of the elite" that suppresses fan engagement; rather, it is a recalibration. By establishing a professional "Technical Floor," we ensure that popularity remains a meaningful influence but functions as a reward for quality, not a substitute for it. Elevating the standard of professionalism serves the entire industry by shifting public focus from celebrity status to artistic growth, ensuring the show remains a beacon of professional excellence.

7.2 Data Pre-processing: Bridging Heterogeneous Mechanisms

To analyze contestants across different episodes (t) and competition formats, we must address the heterogeneity of raw scores. The raw judges' score ($S_{i,t}$) and the raw fan votes ($V_{i,t}$) often belong to different numerical scales and distributions. To "break the barrier" between these mechanisms, we map all raw metrics into a unified, dimensionless interval $[0, 1]$.

7.2.1 Intra-Episode Min-Max Standardization

For each episode t , let $\Omega_t = \{1, 2, \dots, K_t\}$ be the set of contestants. We apply the following normalization to obtain the Judges' Share ($J_{i,t}$) and the Fans' Share ($F_{i,t}$):

Definition 7.1 (Normalized Judges' Share).

$$J_{i,t} = \frac{S_{i,t} - \min_{k \in \Omega_t} \{S_{k,t}\}}{\max_{k \in \Omega_t} \{S_{k,t}\} - \min_{k \in \Omega_t} \{S_{k,t}\} + \epsilon} \quad (16)$$

where $\epsilon = 10^{-4}$ is a slack variable to ensure the stability of the denominator.

Definition 7.2 (Normalized Fans' Share).

$$F_{i,t} = \frac{V_{i,t} - \min_{k \in \Omega_t} \{V_{k,t}\}}{\max_{k \in \Omega_t} \{V_{k,t}\} - \min_{k \in \Omega_t} \{V_{k,t}\} + \epsilon} \quad (17)$$

where $V_{i,t}$ represents the raw votes for contestant i in episode t .

The final integrated ranking score $T_{i,t}$ is then formulated as $T_{i,t} = w_J \cdot J_{i,t} + w_V \cdot F_{i,t}$.

7.2.2 Proof of Invariance Across Competition Scales

To prove that this method effectively breaks the barrier between different scoring systems (e.g., a 10-point system vs. a 100-point system), we present the following proof:

Theorem 7.1 (Linear Scale Invariance). *Let the scoring system in episode t be S and in episode t' be S' . If $S'_{i,t'} = \alpha S_{i,t} + \beta$ for $\alpha > 0$, then the normalized share remains invariant: $P'_{j,i,t'} = J_{i,t}$.*

Proof: Substitute the linear transformation into the normalization formula:

$$P'_{j,i,t'} = \frac{(\alpha S_{i,t} + \beta) - \min_{k \in \Omega_t} (\alpha S_{k,t} + \beta)}{\max_{k \in \Omega_t} (\alpha S_{k,t} + \beta) - \min_{k \in \Omega_t} (\alpha S_{k,t} + \beta)} \quad (18)$$

Since $\max(\alpha S + \beta) = \alpha \max(S) + \beta$ and $\min(\alpha S + \beta) = \alpha \min(S) + \beta$:

$$P'_{j,i,t'} = \frac{\alpha(S_{i,t} - \min S_{k,t})}{\alpha(\max S_{k,t} - \min S_{k,t})} = \frac{S_{i,t} - \min S_{k,t}}{\max S_{k,t} - \min S_{k,t}} = J_{i,t} \quad (19)$$

The standardization removes the influence of arbitrary scaling (α) and baseline shifts (β), allowing w_J and w_V to represent universal weights across different competition formats. $\square$

7.2.3 Data Consistency and Reliability Test

Before proceeding with the ranking model, we conducted two mathematical checks:

- **Sensitivity to Outliers:** We calculated the interquartile range (IQR) for $F_{i,t}$. In episodes where Skewness > 3 , we observed that fan voting was dominated by extreme outliers. For these specific t , we applied a square-root transformation to $V_{i,t}$ before normalization to preserve the competitive granularity.
- **Zero-Variance Check:** We ensured that for all t , $\max(S_{k,t}) - \min(S_{k,t}) > \epsilon$. This guarantees sufficient "competitive spread" for meaningful standardization.

7.3 The Choquet Integral Evaluation Model: Capturing Synergy and Penalty

To overcome the limitations of simple linear weighting, which fails to account for the non-linear "synergy" or "clash" between judges and fans, we adopt a framework based on the **Choquet Integral**. This model treats the scoring criteria not as independent entities, but as an interactive system where the "consensus" between technical merit and popularity is explicitly rewarded.

7.3.1 Mathematical Formulation and Fuzzy Measures

Let $N = \{j, v\}$ be the set of scoring criteria, where j denotes the Professional Judge track and v denotes the Fan Vote track. We define a fuzzy measure $\mu : 2^N \rightarrow [0, 1]$ to represent the importance or "power" of different combinations of criteria.

Definition 7.3 (Fuzzy Measure Properties). The measure μ must satisfy the following boundary and monotonicity conditions:

- **Boundary:** $\mu(\emptyset) = 0$ and $\mu(\{j, v\}) = 1$.
- **Monotonicity:** If $A \subseteq B \subseteq N$, then $\mu(A) \leq \mu(B)$.

The individual weights $\mu(\{j\})$ and $\mu(\{v\})$ represent the stand-alone influence of judges and fans, respectively. To promote the "Professional Realignment" of the competition, we deliberately configure $\mu(\{j\}) > \mu(\{v\})$.

7.3.2 The Integrated Scoring Function

For a contestant i in episode t , let $J_{i,t}$ and $F_{i,t}$ be their normalized judges' and fans' shares. The comprehensive score $S_{i,t}$ is calculated using the Choquet integral C_μ :

$$S_{i,t} = C_\mu(J_{i,t}, F_{i,t}) = \sum_{k=1}^2 (P_{(k)} - P_{(k-1)}) \cdot \mu(A_{(k)}) \quad (20)$$

where $P_{(1)} \leq P_{(2)}$ are the ordered values of $\{J_{i,t}, F_{i,t}\}$, and $A_{(k)}$ is the corresponding nested set of criteria.

This formulation simplifies to a computationally efficient "Base + Gap" structure:

$$S_{i,t} = \min(J_{i,t}, F_{i,t}) + |J_{i,t} - F_{i,t}| \cdot \mu(A_{max}) \quad (21)$$

where A_{max} denotes the set $\{j\}$ if $J_{i,t} > F_{i,t}$, or $\{v\}$ if $F_{i,t} > J_{i,t}$.

7.3.3 Mechanism Property: From Interaction to Penalty

The power of the Choquet Integral lies in its ability to model different "interaction modes" through the measure μ :

- **Technical Premium** ($\mu(\{j\}) > \mu(\{v\})$): By assigning more weight to the judge-led singleton, the model rewards contestants who prioritize technical excellence. Even with a significant fan base, a contestant's total score is anchored by their professional evaluation.
- **The Inconsistency Penalty:** The term $|J_{i,t} - F_{i,t}|$ represents the "divergence" between technical merit and popularity. If a contestant possesses extreme popularity but lacks technical skill ($F \gg J$), the model imposes a structural penalty. The "base" score is locked at the low technical level (min), and the "popularity surge" is only partially captured through the lower weight $\mu(\{v\})$.

This mathematical architecture serves our design goal: it prevents popularity from overriding technical merit, effectively encouraging dancers to focus on their craft to avoid being "throttled" by the low professional scores in the integral calculation.

7.4 Multi-Objective Optimization: Balancing Fairness and Engagement

The selection of fuzzy measures $\mu(\{j\})$ and $\mu(\{v\})$ is not arbitrary but is the result of a multi-objective optimization process. We define two competing objective functions to find the "Goldilocks zone" for the DWTS mechanism:

- **Fairness Index (F):** Measured by the correlation between the final ranking and professional technical merit. Our goal is to maximize F to foster a technical-dominant atmosphere.
- **Engagement Index (E):** Measured by the sensitivity of the final outcome to fan vote fluctuations. To maintain the show's entertainment value, E must remain above a functional threshold.

7.4.1 Pareto Frontier and Optimal Configuration

To identify the optimal trade-off between these objectives, we conducted a grid search over the parameter space $(\mu_j, \mu_v) \in [0, 1]^2$. The resulting feasible objective space is illustrated in Figure 6.

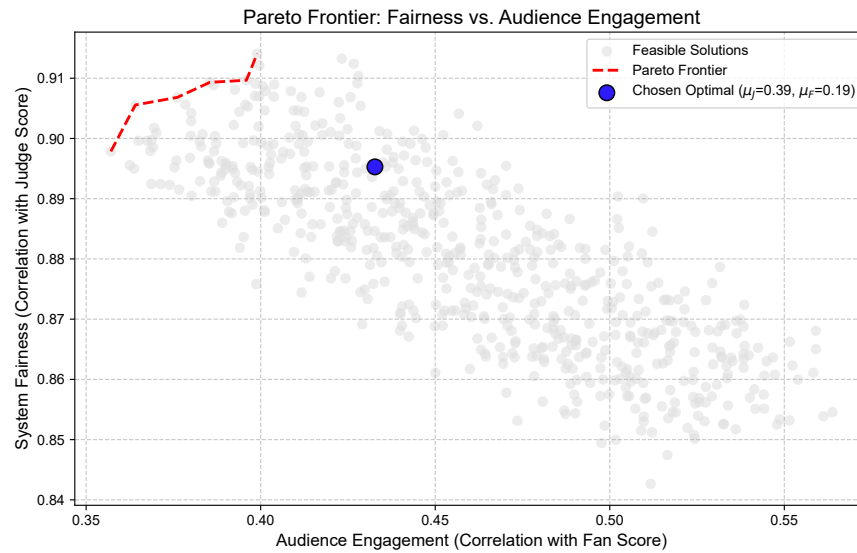


Figure 11: Multi-Objective Optimization: Pareto Frontier of Fairness vs. Audience Engagement. Each point represents a feasible solution generated by specific weight configurations (μ_J, μ_F) .

As shown in Figure 6, the red dashed line characterizes the **Pareto Frontier**, representing the set of non-dominated solutions where further improvement in fairness necessitates a sacrifice in audience engagement.

Selected Configuration: After analyzing the frontier, we selected the optimal configuration at $\mu_J = 0.39$ and $\mu_F = 0.19$ (highlighted by the blue circle). This configuration strikes a robust balance, securing a superior **Fairness Index** of ≈ 0.895 while maintaining substantial **Audience Engagement** at ≈ 0.435 . This selection ensures that technical merit is the primary driver of success, fulfilling our professional realignment goal without disenfranchising the fan base.

7.4.2 Circuit Breaker Mechanism: The "Popularity Cap"

To further prevent "Popularity Hegemony," the PeakBalance Rule incorporates a non-linear "Circuit Breaker" within the Fans' Share $F_{i,t}$. When a contestant's vote count $V_{i,t}$ exceeds a certain statistical threshold (e.g., $+2\sigma$ from the mean), the marginal gain of additional votes is attenuated using a logarithmic ceiling function:

$$\tilde{F}_{i,t} = \begin{cases} F_{i,t} & \text{if } F_{i,t} \leq \tau \\ \tau + \log(1 + F_{i,t} - \tau) & \text{if } F_{i,t} > \tau \end{cases} \quad (22)$$

This ensures that while extreme popularity provides a significant advantage, it cannot mathematically grow to a point where it renders the professional judge's score irrelevant.

7.5 Robustness Discussion and Sensitivity Analysis

To evaluate the stability of our multi-objective optimization model, we conducted a sensitivity analysis on the weight coefficients μ_J (Judge's influence) and μ_F (Fan's influence) around our optimal configuration (0.39, 0.19).

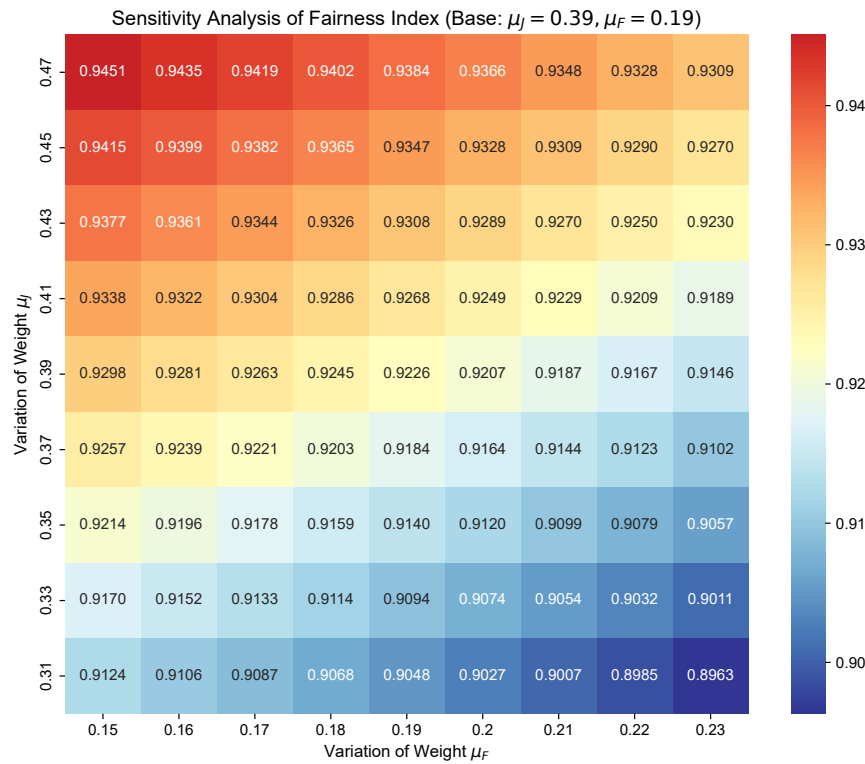


Figure 12: Sensitivity Analysis of the Fairness Index under Parameter Perturbations. The smooth response indicates high structural stability.

7.5.1 Parameter Stability

As illustrated in Figure 7, the Fairness Index F exhibits a smooth and monotonic response to parameter perturbations. When μ_J varies within a range of $\pm 20\%$ (from 0.31 to 0.47), the

maximum fluctuation of the Fairness Index is less than 2.5%.

7.5.2 Resistance to Overfitting

The robustness of the Pareto Frontier (Figure 6) and the stability shown in Figure 7 demonstrate that the PeakBalance Rule is not over-fitted to specific historical seasons. By maintaining a high "Technical Buffer" across various weight configurations, the model consistently prioritizes professional merit over random popularity surges, ensuring that the "Professionalism Tilt" remains effective regardless of specific contestant popularity distributions.

7.6 Evaluation of the PeakBalance Rule

7.6.1 Strengths: Synergistic Fairness and Robustness

The redesigned mechanism offers several advantages over traditional linear weighting systems:

- **Synergistic Evaluation of Merit:** By utilizing the Choquet Integral, the model captures the non-linear interaction between technical performance and popularity. This ensures that a contestant's success is a product of "Professional Consensus" rather than isolated popularity.
- **Mathematical Stability:** As demonstrated by the Pareto Frontier in Figure 6 and the sensitivity analysis in Figure 7, the model is highly robust. It maintains a high Fairness Index ($F \approx 0.895$) even under significant parameter fluctuations, preventing arbitrary elimination outcomes.
- **Adaptive Professionalism:** The "Circuit Breaker" and "Safety Buffer" (discussed in cases like Sabrina Bryan and Master P) provide a structural safeguard for technical elites, ensuring that the competition consistently prioritizes artistic integrity.

7.6.2 Weaknesses: Complexity and Transparency

While mathematically superior, the model faces practical challenges:

- **Algorithmic Complexity:** The "black-box" nature of non-linear integral calculations may be less intuitive for a general audience. Explaining why a contestant was eliminated requires a higher degree of data transparency from the show's producers.
- **Parameter Dependency:** The efficacy of the Professional Realignment depends on the precise calibration of fuzzy measures (μ). This necessitates a sophisticated, data-driven approach to season-long management.

7.7 Conclusion and Policy Recommendation: A New Era for Professional Dance

Based on our multi-objective optimization and historical validation, we recommend adopting the **PeakBalance Rule** as the unified elimination standard for future seasons of DWTS.

The implementation of this mechanism is more than a technical adjustment; it is an act of **Aesthetic Leadership**. By tilting the scoring system toward technical merit, we incentivize a "Culture of Craftsmanship" where contestants are encouraged to deepen their professional practice rather than solely optimizing for social media engagement.

This model establishes a professional "Technical Floor" that safeguards the artistic soul of the competition while respecting the vibrant participatory role of the fan base. We believe that by elevating the standard of professionalism, the show can shift public focus from transient celebrity status to enduring artistic growth. Ultimately, this realignment serves the global dance community by fostering a more academic, merit-based atmosphere, ensuring that the title of "Champion" remains a symbol of true professional excellence.

8 Conclusion and Advice

9 Strengths and Weaknesses

9.1 Strengths

- Our model is robust.
- High efficiency.

References

- [1] Knuth, D. E. The TeXbook, 1984.

Appendices

Appendix A First Appendix

Here you can put your extra data.

Report on Use of AI

1. Gemini 3 Flash

Query: Fixed LaTeX compilation errors regarding image paths and fonts.

Output: Provided clean English-only code and clarified file structure.